

L2.2 The Mean Value Theorem *Recall: L2.1 Maximum and Minimum Values*

Name _____

Solo – 9 minutes

1. Find the critical numbers of $f(x) = (x - 1)^{2/3}$.

Remember both ways a number can be critical.

2. Find the absolute maximum and absolute minimum values of f on the given interval.

$$f(x) = x^3 - 9x + 2 \quad \text{on } [0, 4]$$

Name the method, then run it. Show the candidate list before you compare values.

3. A car covers 120 miles in 2 hours, so its average speed for the trip was 60 mph. **Must** the car have been going exactly 60 mph at some instant? One sentence, and say why.